

Euclid's Elements — Book XI

Proposition 19

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

If two planes cutting one another be at right angles to any plane, then their common section will also be at right angles to the same plane.

1 Introduction

Proposition XI.19 is one of the first places in Book XI where the geometry of three planes must be separated carefully from the diagram that traditionally accompanies the proposition. Euclid's conclusion is metric—a line is perpendicular to a plane—but the most delicate formal step is not metric. It is the incidence assertion that the two traces of the perpendicular planes on the reference plane actually meet.

After the Wyler refactor, this distinction is explicit. The proof is divided into two layers:

1. a neutral common-point core, where a point $D \in \rho \cap \sigma \cap \pi$ is already available;
2. a Euclidean incidence-existence layer proving that such a point must exist from the full hypotheses of XI.19.

The production file is

```
CGJteamLab/Wyler/Proposition11_19.lean
```

with the main theorem

```
euclid_proposition_11_19_wyler
```

The neutral core is isolated in

```
CGJteamLab/Wyler/Proposition11_19_common_point_wyler.lean
```

and the local normal-extraction infrastructure is in

```
CGJteamLab/Wyler/HilbertWylerPlaneNormal.lean
```

2 Euclid's Statement

Euclid states Proposition XI.19 as follows:

If two planes cutting one another be at right angles to any plane, then their common section will also be at right angles to the same plane.

Let ρ and σ be two distinct planes which cut one another, and let π be a third plane. The hypotheses are

$$\rho \perp \pi, \quad \sigma \perp \pi.$$

If

$$t = \rho \cap \sigma,$$

then the conclusion is

$$t \perp \pi.$$

In the formal reconstruction the phrase “common section” is represented by an exact line-carrier description of the intersection of the two planes, not merely by the existence of one line contained in both.

3 Classical Configuration

The classical configuration consists of three planes:

- the reference plane π ;
- a plane ρ perpendicular to π ;
- a plane σ perpendicular to π .

The planes ρ and σ cut one another in a line t . Their traces on π are

$$r = \rho \cap \pi, \quad s = \sigma \cap \pi.$$

The essential point of the formal proof is that the intersection point

$$D = r \cap s$$

must first be obtained. Once it is known, one has

$$D \in \rho \cap \sigma \cap \pi,$$

and the remaining argument is neutral.

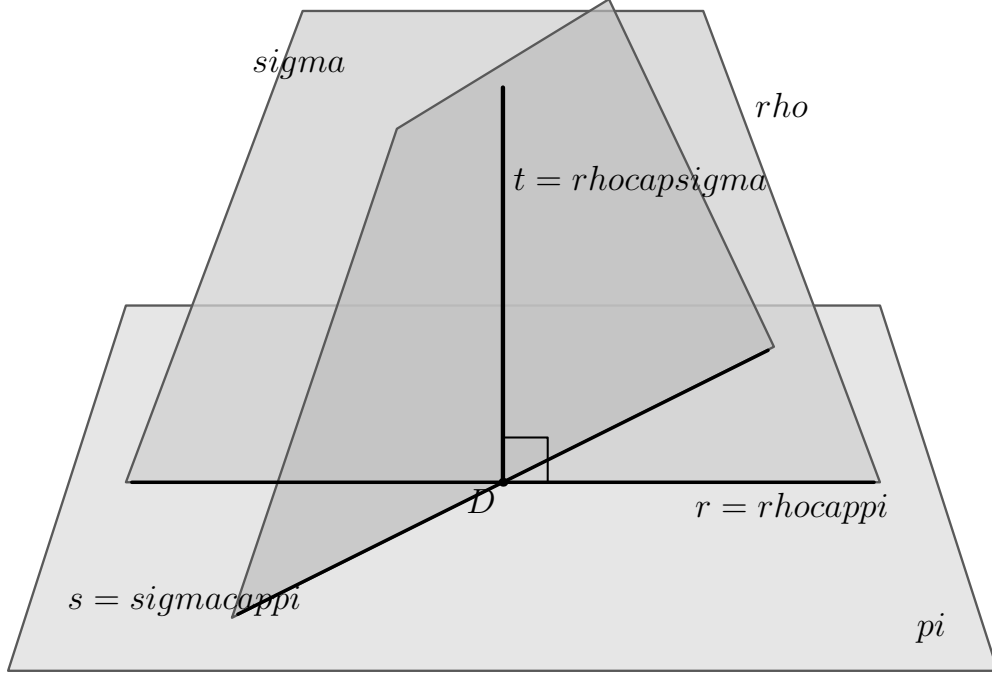


Figure 1: The classical configuration of Proposition XI.19. The planes ρ and σ are both perpendicular to the reference plane π . Their traces $r = \rho \cap \pi$ and $s = \sigma \cap \pi$ meet at D , and the common section $t = \rho \cap \sigma$ passes through the same point. The conclusion is $t \perp \pi$.

4 Euclid's Classical Proof

Euclid lets the two given perpendicular planes cut one another in the line AB . He chooses a point D on the common section at which the relevant lines in the two planes meet the reference plane. In the two given planes he constructs lines through D perpendicular to the corresponding traces in the reference plane.

Because each of the two planes is perpendicular to the reference plane, Euclid's definition of perpendicular planes implies that these constructed lines are perpendicular to the reference plane. Since two straight lines cannot both be perpendicular to the same plane at the same point unless they coincide, the two constructed lines are one and the same line. This common line is then the common section of the two planes and is therefore perpendicular to the reference plane.

The printed proof treats the existence of the point D as part of the configuration. In a formal proof this is the main hidden incidence datum: one must show that the two traces of the given perpendicular planes on π meet.

5 Classical Dependency Pattern

At the level of the traditional diagram the proof has the pattern

$$\begin{array}{c}
\rho \perp \pi, \quad \sigma \perp \pi, \quad \rho \cap \sigma \neq \emptyset \\
\Downarrow \\
D \in \rho \cap \sigma \cap \pi \\
\Downarrow \\
\text{construct } g \subset \rho \text{ and } h \subset \sigma \text{ through } D \\
\Downarrow \\
g \perp \pi, \quad h \perp \pi \\
\Downarrow \\
\text{uniqueness of the perpendicular at } D \\
\Downarrow \\
g = h = \rho \cap \sigma \\
\Downarrow \\
(\rho \cap \sigma) \perp \pi.
\end{array}$$

The step producing D is visually immediate in the classical figure but is not logically free. The Wyler refactor makes it an explicit theorem-level subproblem.

6 What Is Genuinely Three-Dimensional Here?

XI.19 is genuinely spatial in three separate senses.

First, the hypotheses concern perpendicularity between planes rather than between lines. The project represents this through a directed XI.Def.4 predicate, not by replacing plane perpendicularity with the modern statement that one plane merely contains some normal to the other.

Second, the proof uses exact plane intersections. The objects

$$r = \rho \cap \pi, \quad s = \sigma \cap \pi, \quad t = \rho \cap \sigma$$

are ambient spatial lines obtained from plane meets.

Third, the existence of the triple point

$$D \in \rho \cap \sigma \cap \pi$$

requires a spatial argument. If the two traces $r, s \subset \pi$ were parallel, normals erected in ρ and σ would give a second pair of parallel directions. XI.15 would then force $\rho \parallel \sigma$, contradicting the hypothesis that the two planes cut one another.

Thus the central nontriviality of XI.19 is not merely a right-angle calculation. It is the interaction between plane meets, line parallelism, and uniqueness of a plane normal.

7 Formal Statement

The production theorem is:

```

theorem euclid_proposition_11_19_wyler
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  [HSE : HilbertSpaceEuclidean Geo]
  (rho sigma pi : S.Plane)
  (K : Geo.Point)
  (hRhoSigma : Ne rho sigma)
  (hKrho : S.OnPlane K rho)
  (hKsigma : S.OnPlane K sigma)
  (hRhoPerp :
    HilbertSpacePlanesPerpendicular Geo rho pi)
  (hSigmaPerp :
    HilbertSpacePlanesPerpendicular Geo sigma pi) :
  exists D : Geo.Point,
  exists t : Geo.Line,
  S.OnPlane D pi /\
  H.OnLine D t /\
  HilbertLineInPlane Geo t rho /\
  HilbertLineInPlane Geo t sigma /\
  (forall X : Geo.Point,
    (S.OnPlane X rho /\ S.OnPlane X sigma) <=>
      H.OnLine X t) /\
  HilbertLinePerpendicularPlaneAt Geo t pi D

```

The point K records explicitly that ρ and σ have a common point. Together with $\rho \neq \sigma$, this is the formal reading of Euclid’s phrase “two planes cutting one another.”

The result does more than assert the existence of a normal line. It returns a line t together with the exact intersection property

```

forall X : Geo.Point,
  (S.OnPlane X rho /\ S.OnPlane X sigma) <=> H.OnLine X t

```

so that t is certified as the common section itself.

8 Hilbert–Wyler Architecture

The proof uses three distinct layers of the project architecture.

8.1 Directed perpendicular-plane structure

The predicate

```
HilbertSpacePlanesPerpendicular Geo rho pi
```

is built from

```
HilbertSpacePlanesPerpendicularAlong Geo rho pi s
```

which records:

- the exact common section s of the two planes;
- the inclusion $s \subset \rho$ and $s \subset \pi$;
- the XI.Def.4 condition that every line of ρ perpendicular to s at a section point is perpendicular to the whole plane π .

This prevents XI.18 and XI.19 from becoming tautological by definition.

8.2 Normal extraction

The helper

```
hilbertSpacePlanesPerpendicularAlong_normal_exists_wyler
```

states that at any point D of the common section one may erect inside the perpendicular plane a line perpendicular to the reference plane.

Its construction is planar: it enters **PlaneGeo** of the carrier plane, erects a right angle to the section line, and transports the resulting perpendicularity back to the ambient geometry.

8.3 Exact plane meets

The proof repeatedly uses the spatial intersection-line API to identify the full carrier of a plane intersection. Exactness is essential in the neutral core: after XI.13 identifies two candidate normals, their common containment in ρ and σ forces that normal to equal the actual meet line.

9 Ambient Space and PlaneGeo

The public statement and the global proof remain in the ambient geometry **Geo**. The principal objects are

```
rho sigma pi : S.Plane  
r s t        : Geo.Line  
D K          : Geo.Point
```

The induced planar geometry **PlaneGeo** appears only in the lower normal-construction helper. There, after a point D on a section line is fixed, an ordinary planar perpendicular line is erected inside the carrier plane.

This division is important. The following tasks are ambient and spatial:

- intersecting two planes;
- proving two ambient lines spatially parallel;
- proving two ambient planes parallel;
- transporting line–plane perpendicularity.

Only the local construction “erect a perpendicular to a line at a point in one plane” belongs naturally to **PlaneGeo**.

10 Lean Proof Architecture

The production proof has two major stages.

10.1 Stage 1: force the triple point

Let

$$r = \rho \cap \pi, \quad s = \sigma \cap \pi.$$

Assume for contradiction that r and s do not meet. Since both lie in π , they are spatially parallel:

$$r \parallel s.$$

Choose $R \in r$ and $Q \in s$. XI.Def.4 gives normals

$$g \subset \rho, \quad g \perp \pi \text{ at } R,$$

and

$$h \subset \sigma, \quad h \perp \pi \text{ at } Q.$$

The disjointness of r and s implies $R \neq Q$. Hence XI.6 gives

$$g \parallel h.$$

The pairs (r, g) and (s, h) are intersecting pairs in ρ and σ , respectively. They cannot all lie in one plane, for otherwise the first pair would identify that plane with ρ , while the second would identify it with σ , contradicting $\rho \neq \sigma$.

Therefore XI.15 applies to the two pairs of parallel lines

$$r \parallel s, \quad g \parallel h,$$

and yields

$$\rho \parallel \sigma.$$

But $K \in \rho \cap \sigma$, contradiction. Thus r and s meet at a point D , and

$$D \in \rho \cap \sigma \cap \pi.$$

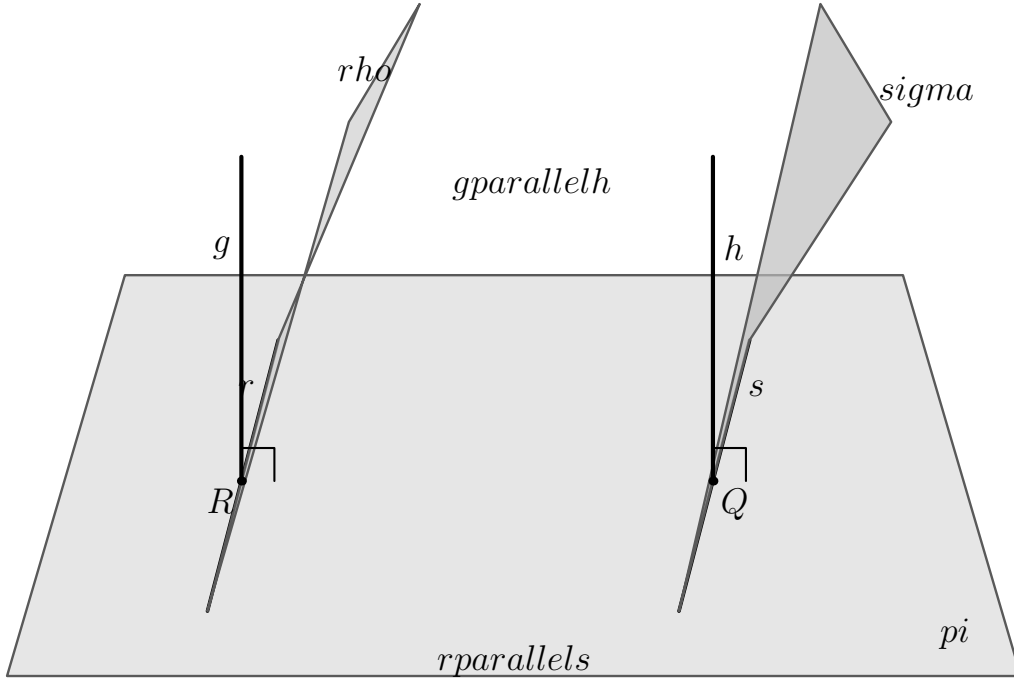


Figure 2: The formal incidence step suppressed by the classical picture. If the traces $r = \rho \cap \pi$ and $s = \sigma \cap \pi$ were disjoint, then $r \parallel s$. Normals g and h to π , erected in ρ and σ , are parallel by XI.6. XI.15 would then force $\rho \parallel \sigma$, contradicting the given common point of ρ and σ . Therefore the traces meet at the required triple point D .

10.2 Stage 2: apply the neutral common-point core

At the point D , XI.Def.4 supplies

$$g \subset \rho, \quad g \perp \pi \text{ at } D,$$

and

$$h \subset \sigma, \quad h \perp \pi \text{ at } D.$$

XI.13 gives

$$g = h.$$

Thus one line perpendicular to π at D lies in both ρ and σ . The exact meet of those two planes is a line t through D . A second point on g , together with exact meet membership, gives

$$g = t.$$

Therefore

$$t \perp \pi.$$

This second stage is exactly

```
euclid_proposition_11_19_common_point_wyler
```

and is neutral.

11 Wyler-Style Flat Reconstruction

The Wyler-style contribution to XI.19 is primarily organizational rather than a replacement of the metric content. Plane intersections are treated as exact rank-two meets with rank-one carriers, while the proof keeps the metric notion of perpendicularity separate.

The three relevant meet lines are

$$\text{carrier}(r) = \text{carrier}(\rho) \cap \text{carrier}(\pi),$$

$$\text{carrier}(s) = \text{carrier}(\sigma) \cap \text{carrier}(\pi),$$

and

$$\text{carrier}(t) = \text{carrier}(\rho) \cap \text{carrier}(\sigma).$$

The proof first establishes that the two rank-one traces r, s in the rank-two flat π have a common point. Once that point is known, the metric part identifies the rank-one meet t with the unique normal to π at that point.

Thus XI.19 is a useful example of the intended division of labor:

- flat incidence determines where the relevant carriers meet;
- Hilbert congruence/right-angle structure supplies normality;
- uniqueness identifies the metric line with the flat-theoretic meet.

The flat language therefore clarifies the configuration without erasing the metric theorem being proved.

12 Hidden Formal Data

Several pieces of information hidden in the classical picture become explicit in Lean.

12.1 The cutting planes are distinct

The hypothesis

```
hRhoSigma : Ne rho sigma
```

is necessary. Without it the phrase “their common section” would not determine a line.

12.2 The cutting planes really have a common point

The point

```
K : Geo.Point  
hKrho   : S.OnPlane K rho  
hKsigma : S.OnPlane K sigma
```

formalizes the statement that ρ and σ cut one another.

12.3 The triple point is not an input

The point D satisfying

$$D \in \rho \cap \sigma \cap \pi$$

is not assumed. It is constructed by contradiction from XI.6 and XI.15. This is the principal incidence content exposed by the refactor.

12.4 Common sections are exact

For each pair of distinct planes the formal development records not only a line contained in both planes but the equivalence

```
(S.OnPlane X rho /\ S.OnPlane X sigma) <-> H.OnLine X t
```

for every point X . This exactness is what allows the common normal to be identified with the actual intersection line.

13 Relationship with Earlier and Later Book XI Propositions

XI.19 is strongly cumulative. Its refactored proof uses the following previous Book XI results.

XI.6. Two lines perpendicular to the same plane at distinct points are parallel. This is used in the contradiction argument for the traces r and s .

XI.13. The perpendicular to a plane at a prescribed point is unique. This is the central theorem in the neutral common-point core.

XI.15. Two planes carrying two corresponding pairs of parallel lines are parallel, under the required noncoplanarity condition. This is the Euclidean step that forces the two traces on π to meet.

XI.18 and XI.Def.4 infrastructure. The refactor preceding XI.19 introduced a faithful directed predicate for perpendicular planes and the elimination rule saying that a line in the first plane perpendicular to the common section is perpendicular to the second plane. XI.19 relies on that infrastructure rather than introducing a new ad hoc notion of plane perpendicularity.

The proposition in turn supplies a reusable principle for later arguments in which a line defined as the intersection of two perpendicular carrier planes must be recognized as a normal to a reference plane.

14 Relationship with Hilbert's Axiom Groups

The dependency split is especially informative.

The neutral common-point core uses:

- Group I spatial incidence for lines and planes;
- Group II order through the planar right-angle infrastructure;
- Group III congruence for perpendicularity and uniqueness of normals.

It does not require the Euclidean parallel axiom.

The full production theorem additionally requires

```
HilbertSpaceEuclidean Geo
```

because its triple-point existence argument invokes XI.15. Thus Group IV is not part of the local metric conclusion. It enters only in proving that the configuration required by the neutral core must exist globally.

This separation is one of the main proof-theoretic outcomes of the refactor.

15 Classical DAG versus Formal DAG

The central difference is the treatment of the triple point.

Euclid.

$$\begin{array}{c}
\rho \perp \pi, \quad \sigma \perp \pi, \quad \rho \text{ cuts } \sigma \\
\Downarrow \\
\text{diagram contains } D \in \rho \cap \sigma \cap \pi \\
\Downarrow \\
\text{two normals to } \pi \text{ at } D \\
\Downarrow \\
\text{uniqueness} \\
\Downarrow \\
(\rho \cap \sigma) \perp \pi.
\end{array}$$

Lean / Hilbert–Wyler.

$$\begin{array}{c}
\rho \perp \pi, \quad \sigma \perp \pi, \quad K \in \rho \cap \sigma, \quad \rho \neq \sigma \\
\Downarrow \\
r = \rho \cap \pi, \quad s = \sigma \cap \pi \\
\Downarrow \\
\text{assume } r \cap s = \emptyset \\
\Downarrow \\
r \parallel s \\
\Downarrow \\
\text{XI.Def.4 gives normals } g, h \text{ to } \pi \\
\Downarrow \\
\text{XI.6 : } g \parallel h \\
\Downarrow \\
\text{XI.15 : } \rho \parallel \sigma \\
\Downarrow \\
\perp \text{ from } K \in \rho \cap \sigma \\
\Downarrow \\
\exists D \in \rho \cap \sigma \cap \pi \\
\Downarrow \\
\text{XI.Def.4 + XI.13} \\
\Downarrow \\
(\rho \cap \sigma) \perp \pi.
\end{array}$$

The formal DAG is longer, but the additional nodes do not represent bureaucratic Lean work. They expose a genuine incidence dependency hidden by the classical diagram.

16 Structural Lesson

XI.19 shows why the Book XI refactor must keep three notions distinct:

1. exact flat intersection;
2. perpendicularity of planes in the sense of XI.Def.4;
3. existence and uniqueness of a line normal to a plane at a point.

Collapsing plane perpendicularity to the modern existential statement “the first plane contains a normal to the second” would make the later logic much less informative. In particular, XI.18

would become close to a definitional tautology and XI.19 would lose the visible role of the common section.

The current architecture instead lets XI.Def.4 act as an elimination rule: from a perpendicular-plane certificate and a line perpendicular to the common section, one obtains line-plane perpendicularity. XI.13 then supplies the uniqueness needed to identify normal directions coming from different planes.

The resulting proof is a good model for the Wyler program: first expose the flat configuration exactly, then add only the metric information required by the proposition.

17 Conclusion

Proposition XI.19 has the compact classical conclusion

$$\rho \perp \pi, \quad \sigma \perp \pi, \quad \rho \text{ cuts } \sigma \quad \Longrightarrow \quad (\rho \cap \sigma) \perp \pi.$$

The formal reconstruction reveals that this statement contains two logically different problems.

The first is global and incidence-theoretic: the traces of ρ and σ on π must meet. The production proof derives this by contradiction using XI.6 and XI.15, and this is the only stage requiring the Euclidean spatial parallel interface.

The second is local and neutral. At the resulting triple point, XI.Def.4 produces a normal to π in each of the two planes, XI.13 identifies those normals, and exact plane intersection identifies the common normal with the common section of ρ and σ .

After the Wyler refactor, XI.19 therefore becomes a particularly clear example of how incidence, flat structure, and metric structure should be layered rather than conflated in the formalization of Euclid's solid geometry.